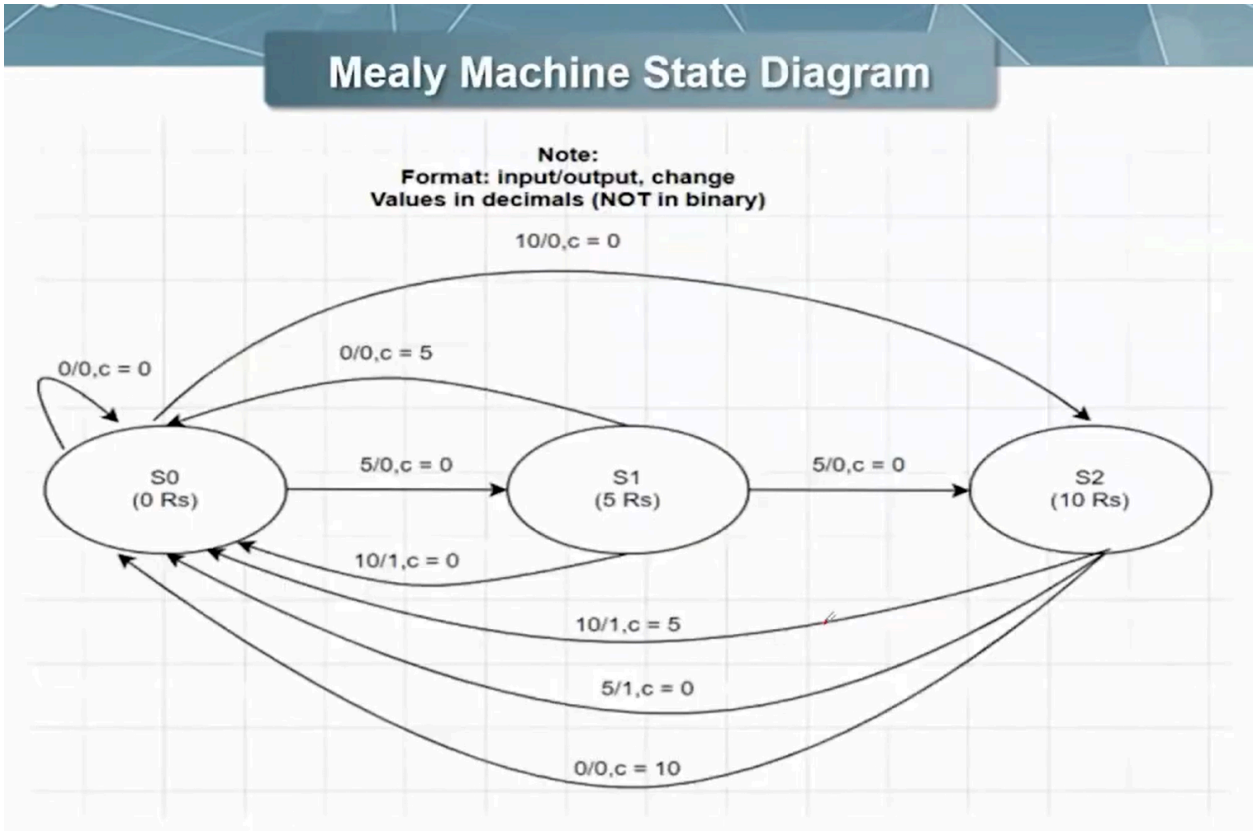
VENDING MACHINE

A vending machine is implemented in Verilog using a finite state machine (FSM) to simulate the process of accepting coins and dispensing a product when the required amount is inserted. The system transitions between states based on the input coins and resets after delivering the item.



This is a **Mealy Finite State Machine (FSM)** for a vending machine. In a **Mealy machine**, the **output depends on both the current state and the input coin**.

The vending machine accepts **₹5 and ₹10 coins** and dispenses the product when the required amount is reached. After dispensing, it resets to the initial state.

The states represent the **total money currently inserted**.

State S0 – (0 Rs)

This is the **initial state** of the machine.

- It means **no money has been inserted yet**.
- If the user inserts **₹5**, the machine moves to **State S1 (5 Rs collected)**.
- If the user inserts **₹10**, the machine moves to **State S2 (10 Rs collected)**.
- If **₹0 is inserted**, the machine **remains in S0**.

So S0 basically **waits for the first coin**.

State S1 – (5 Rs)

This state means the machine has **received ₹5** so far.

Possible transitions:

- If another **₹5 coin is inserted**, total becomes **₹10**, so it moves to **State S2**.
- If a **₹10 coin is inserted**, the total becomes **₹15**, which is enough to buy the product.
    - The machine **dispenses the item (output = 1)**
    - Returns to **State S0**.
- If **₹0 is inserted**, the machine **stays in S1** and may return change depending on the design.

So S1 represents **partial payment (₹5 inserted)**.

State S2 – (10 Rs)

This state means the machine has **already collected ₹10**.

Possible transitions:

- If **₹5 coin is inserted**, the total becomes **₹15**, so:

    - The machine **dispenses the product (output = 1)**
    - Returns to **State S0**.
- If **₹10 coin is inserted**, the machine:
    - **dispenses the product**
    - **returns ₹5 change**
    - resets to **S0**.
- If **₹0 is inserted**, it may return change or reset depending on the transition.

So S2 represents **almost complete payment (₹10 inserted)**.